BONE PATHOLOGY III
BONE TUMORS

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I. PRIMARY BONE TUMORS

- **Age**: Most develop during **first decades of life**

Gender: Affect males >females

Location: Most affect **long bones** and most occur in **metaphysis** of long bones

Genetic: Genetic alteration may occur in some**: p53 and RB gene** mutations in **osteosarcoma**

BONE TUMORS

- **Benign** abnormalities of bone are common, especially in **children.**

- **Malignant** bone sarcomas are aggressive and have a poor outcome

History and clinical examination

History:

- **Pain:** unremitting , *especially at night,*
- **Mass**

Past history:

- Paget’s disease of the bone.
- Metastatic disease
- Multiple myeloma.

Gout can mimic osteogenic tumours in presentations ,

History and clinical examination

Physical examination:

- Tenderness, swelling or deformity
- Skin changes
- Masses:
  - A useful rule of thumb is that **masses greater than 5 cm** in maximum dimension and depth to fascia should be considered malignant until proven otherwise.
  - A golf ball is a useful comparator.
- Local lymph nodes

Histologic type	Benign	Malignant
Hematopoietic		Myeloma Malignant lymphoma
Chondrogenic	Osteochondroma Chondroma Chondroblastoma	Chondrosarcoma
Osteogenic	Osteoid osteoma Osteoblastoma	Osteosarcoma
Fibrogenic	Firoma Nonossifying fibroma Fibrous histocytoma	Fibrosarcoma
Unknown	Giant cell tumor	
Neuroectodermal		Ewing tumor
Notochordal	Benign notochordal cell tumor	Chordoma

The more likely differential diagnoses by age

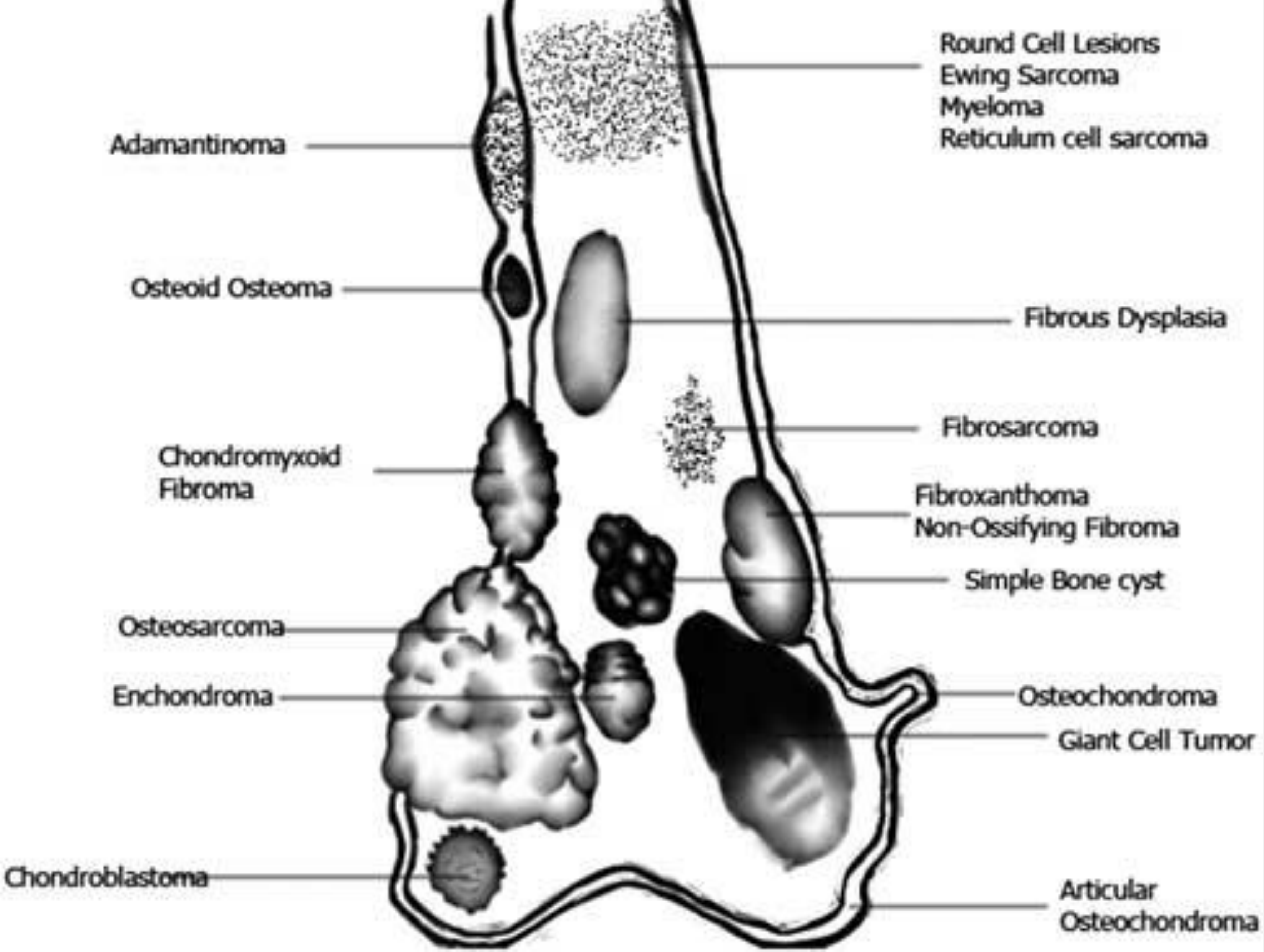
0-5 years	Osteomyelitis Fibrous dysplasia **Metastatic neuroblastoma** Leukemia
5-20 years	Osteomyelitis Fibrous dysplasia **Osteosarcoma** **Ewing's tumour** Leukemia
> 40 years	**Metastatic bone disease** Myeloma Lymphoma Paget's disease Bone infarct **Chondrosarcoma**

Clinical, Radiology and Histology

1. **Where is the lesion?** In which bone and within which anatomical region of that bone?

2. **How is the bone responding?** The universal response of bone to disease or injury is to make more bone. **Osteolytic/Osteoblastic**

3. What is in the lesion (any clues to the histology)? **histological diagnosis**



The common anatomical location of bone tumours

A. **OSTEOGENIC** TUMORS AND TUMOR LIKE LESIONS

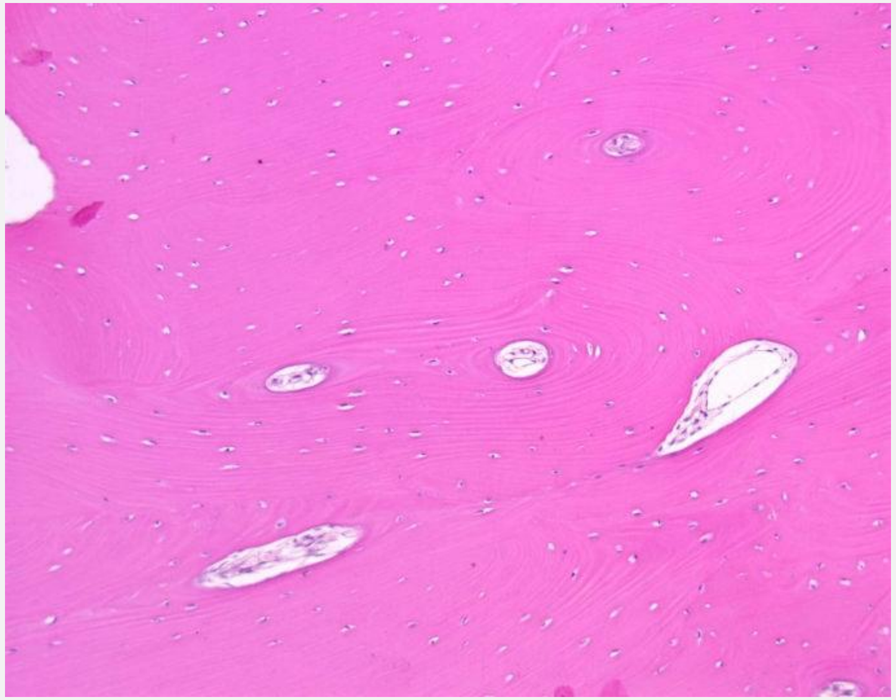
1. Osteoma
2. Osteoid osteoma and Osteoblastoma
3. Osteosarcoma

1. OSTEOMA

- Benign
- Most on **surface of facial bones**
- Associated with **Gardner syndrome-**familial colorectal polyposis-

Osteoma

- Well-demarcated tumors
- Dense compact bone: same density as bone
- Microscopically : normal bone tissue



2. OSTEOID OSTEOMA AND OSTEOBLASTOMA

Osteoid osteoma and osteoblastoma are **benign bone-producing tumors** that have similar histologic features but differ clinically and radiographically.

Osteoid Osteoma

- Benign osteoblast tumour = osteoid
- Surrounded by rim of reactive bone = osteoma

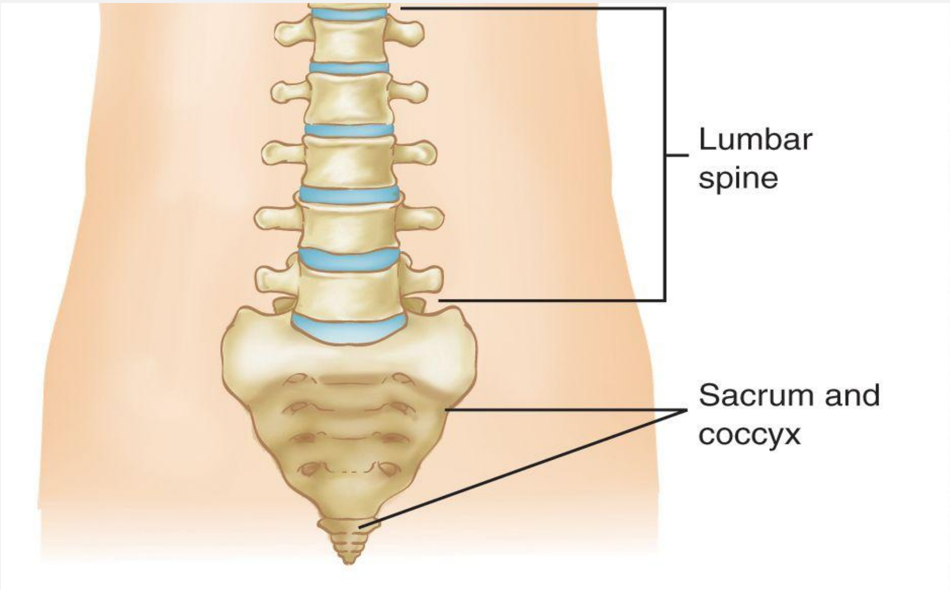
- Young adults < 25yo

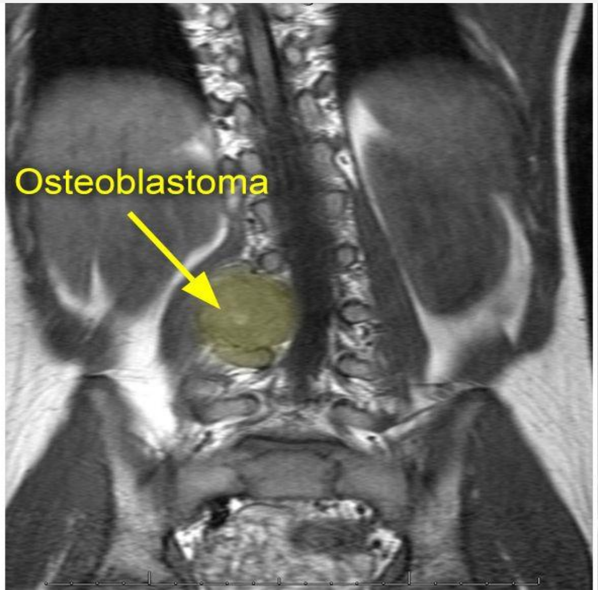
- In **cortex** of **diaphysis** of **long** bones (appendicular)

- Presents
  - **Bone pain** that resolves with **aspirin**
  - Small (<2 cm) boney mass [osteoma] within **radiolucent** [osteoid]

Osteoblastoma

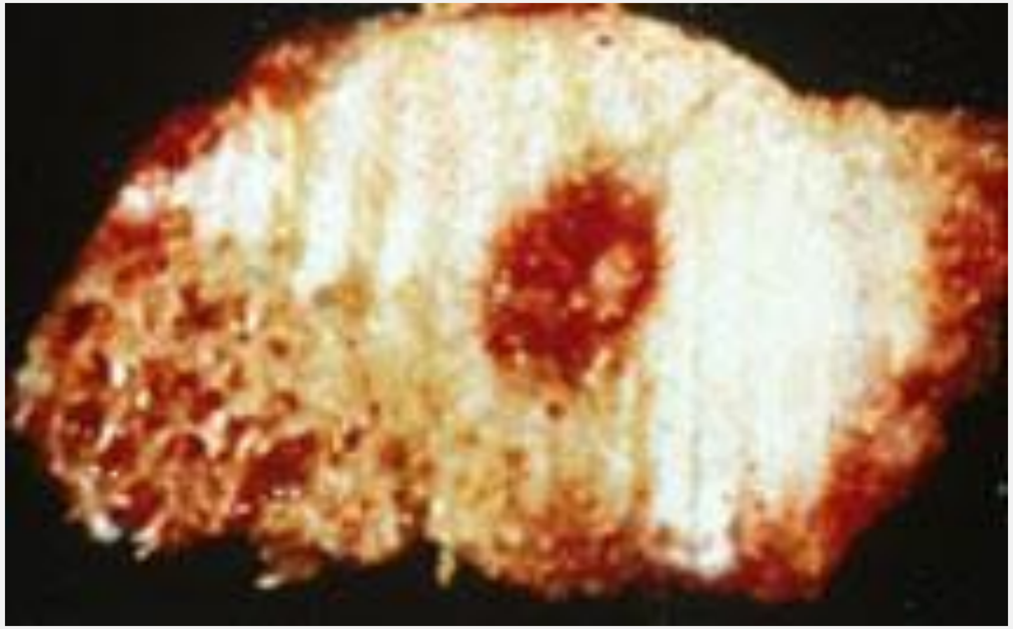
- involves the posterior components of the **vertebrae** (laminae and pedicles) more frequently. (Axial)
- pain is **unresponsive to aspirin,**
- tumor usually **does not induce** a marked bony reaction.



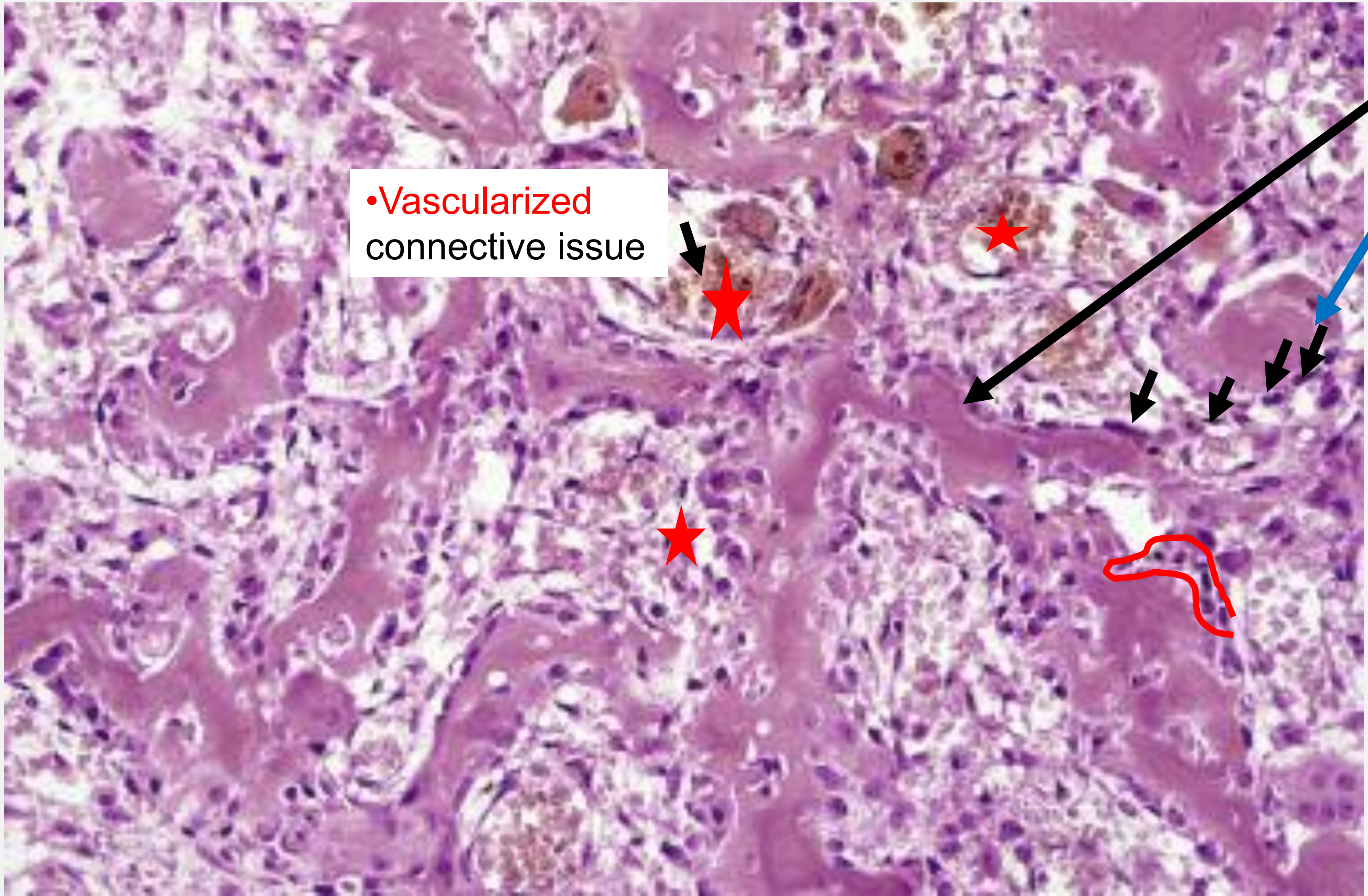


Osteoid Osteoma

- Well circumscribed lesion
- round-to-oval masses of hemorrhagic gritty tan tissue



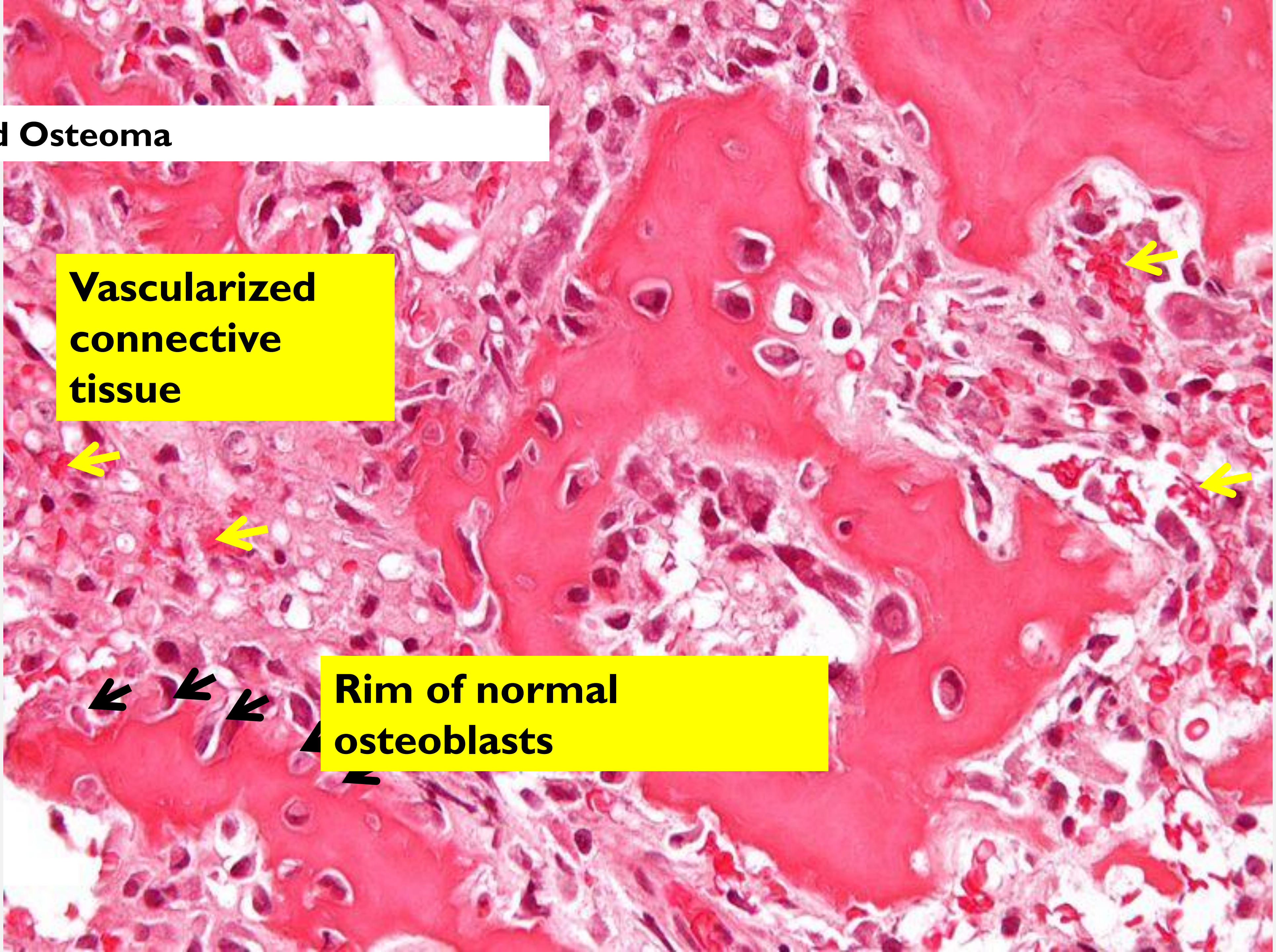
Osteoid Osteoma histology



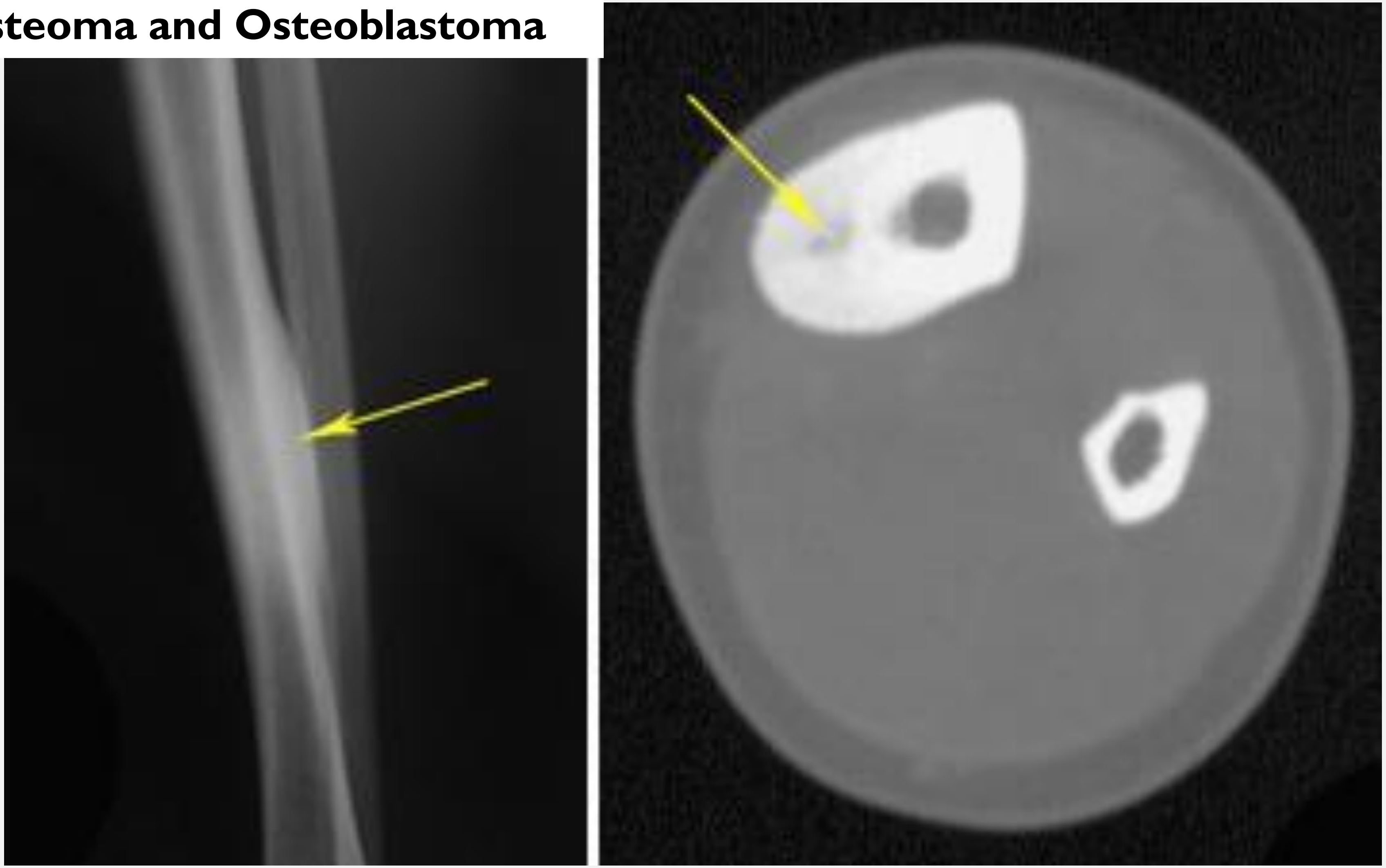
Nidus:

- Haphazard trabeculae
- Rim of normal osteoblasts

Osteoid Osteoma



Osteoid Osteoma and Osteoblastoma

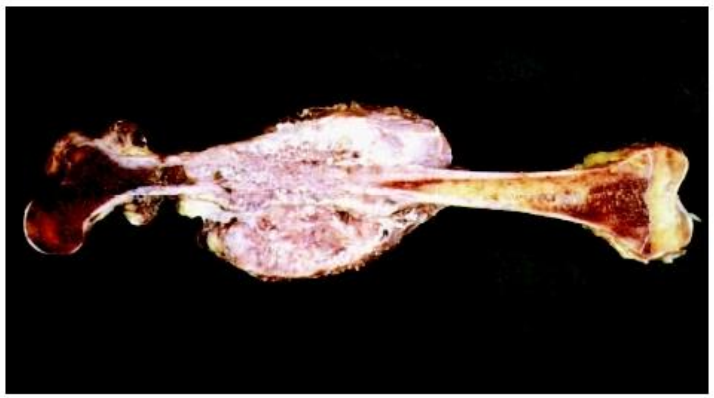


Central translucency and thick cortex

	Osteoid osteoma	Osteoblastoma
Location	Long bones and spine	Axial
Age	Teens	
Clinical features	Nocturnal pain relived by aspirin	Unresponsive to Aspirin
Microscopic	elicit the formation of a large amount of reactive bone, which encircles the lesion.	No marked bone reaction
Size	Less than 2 cm	Greater than 2cm

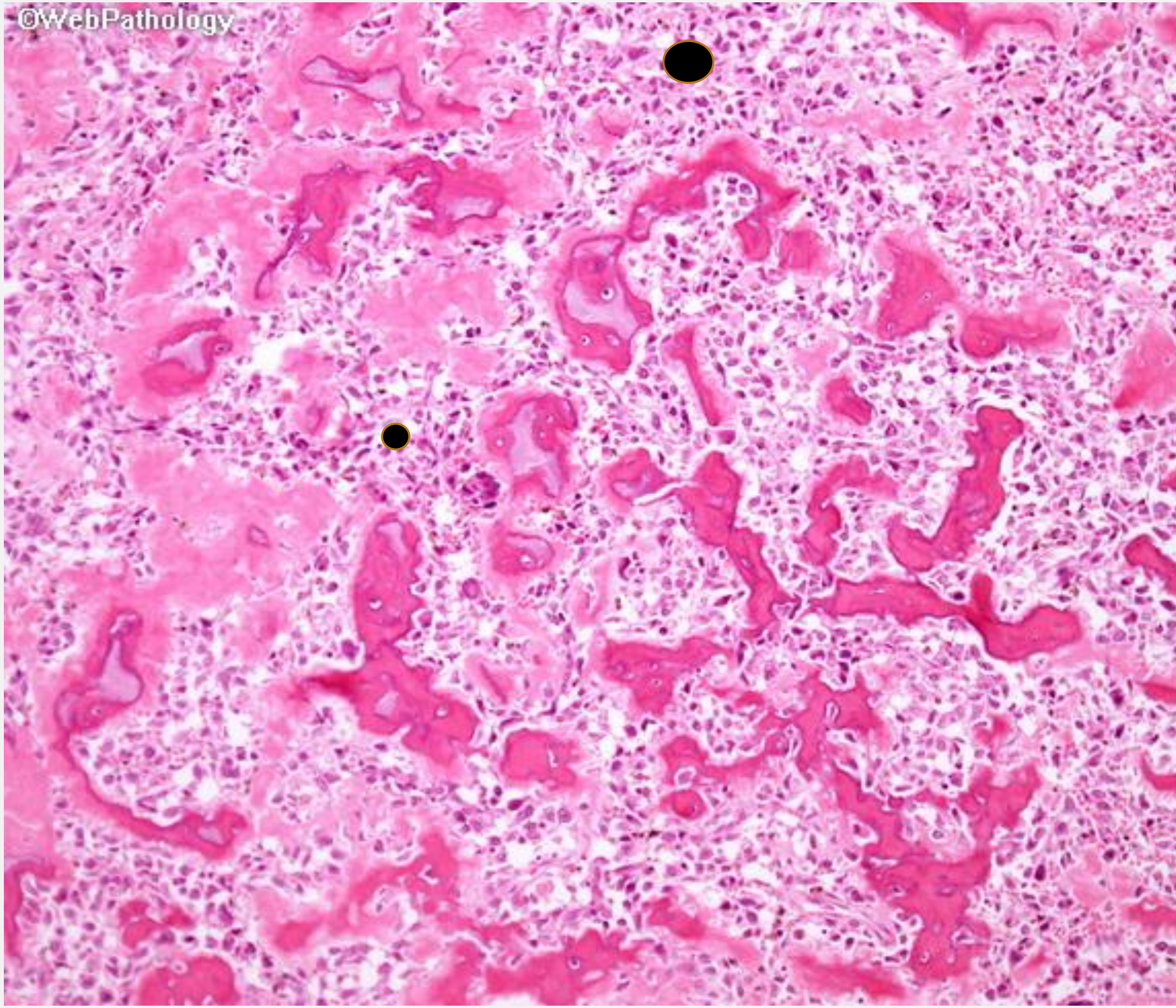
3. OSTEOSARCOMA

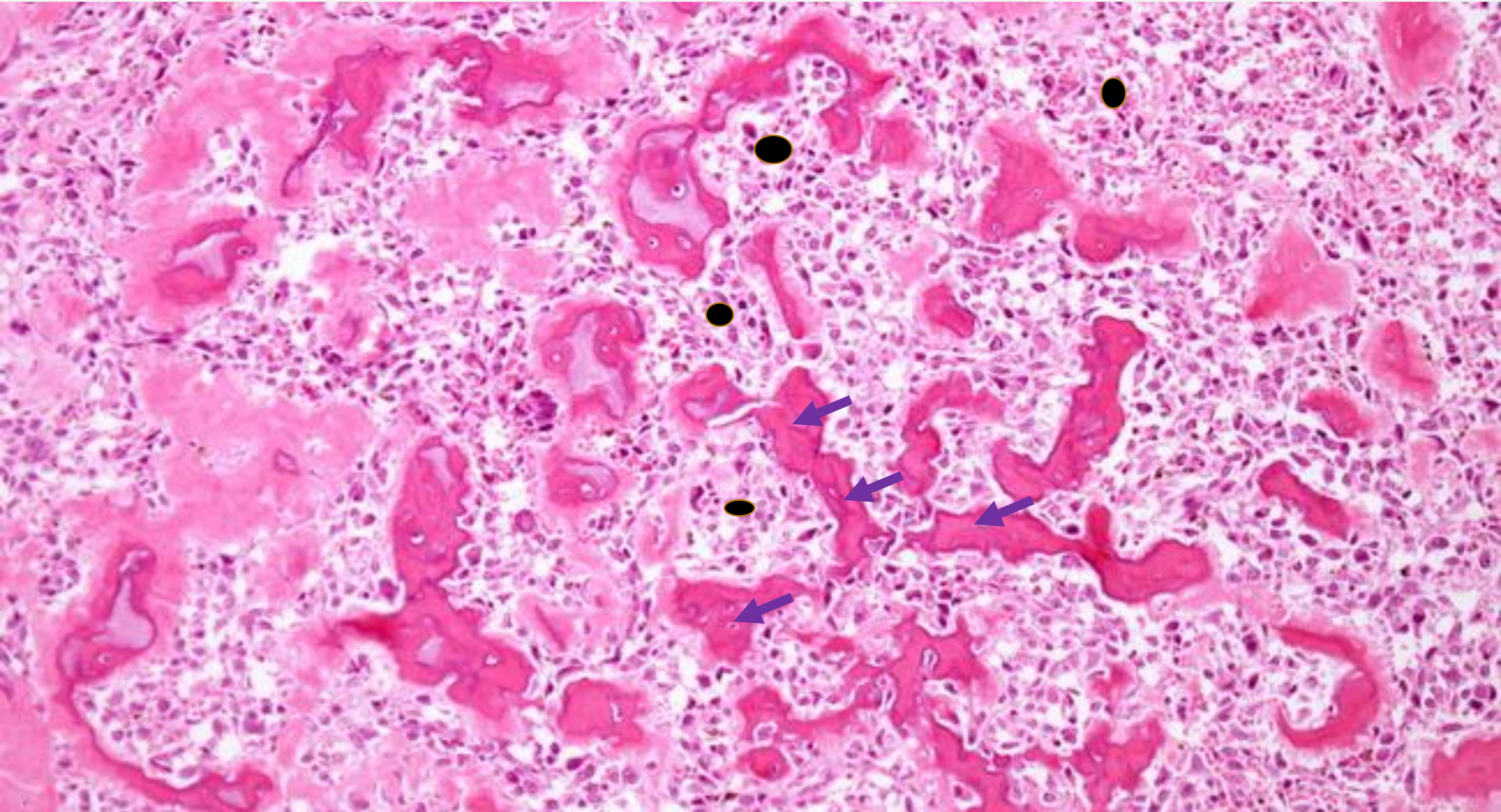
- A case of **17-year-old male**, was admitted for the first time due to **pain and deformity** of the left leg.
- 3 years prior to admission, patient noted a **slowly growing mass** on the left leg which initially was not associated with pain or tenderness and there was no limitation of motion noted.



Osteosarcoma histology

- Spindle cells of mesenchymal origin depositing **immature osteoid matrix**
- **Osteoblastic**
- **anastomosing network of delicate trabeculae** (light pink) some of which are mineralized (dark pink).
- The space between the trabeculae is filled up with **tumor cells (black circles)** with irregular hyperchromatic nuclei



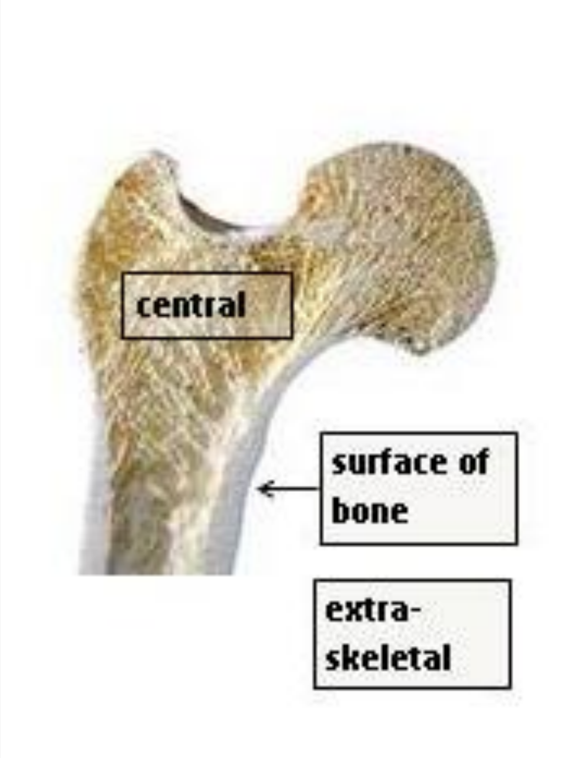


Osteosarcoma

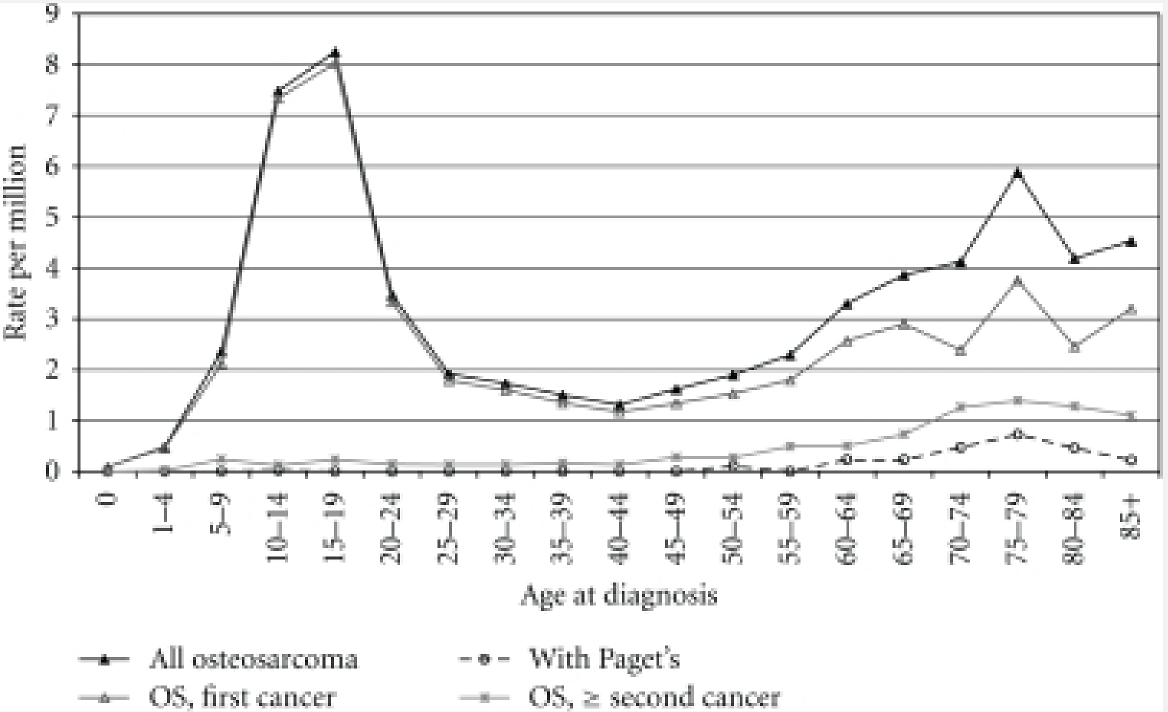
- OS is the **third** most common cancer in **adolescence,** with only **lymphomas and brain tumors** being more prevalent

- 70% of osteosarcomas have acquired **genetic abnormalit**ies, mutation

- 90% are located **centrally** within the bone



Bimodal Distribution of Osteosarcoma



1st peak

- **Long bones**

2nd smaller peak

- **Associated with Paget's, radiation, infarct**
- **Flat bones and long bones**

Presenting signs and evaluation

Symptoms

- **Pain during sleep, enlarging mass**

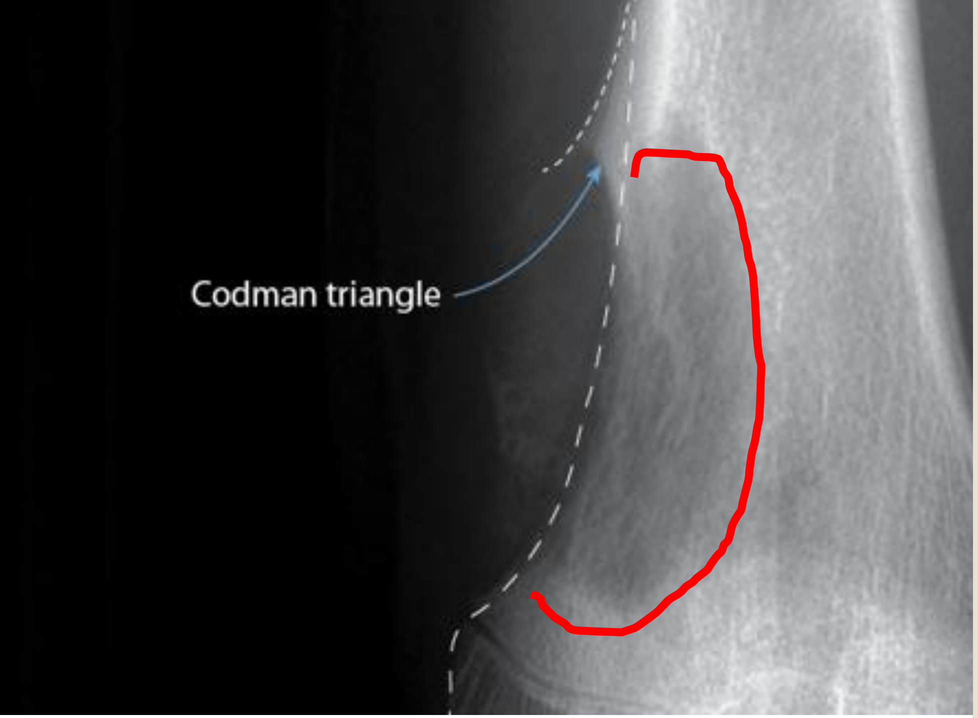
Examination

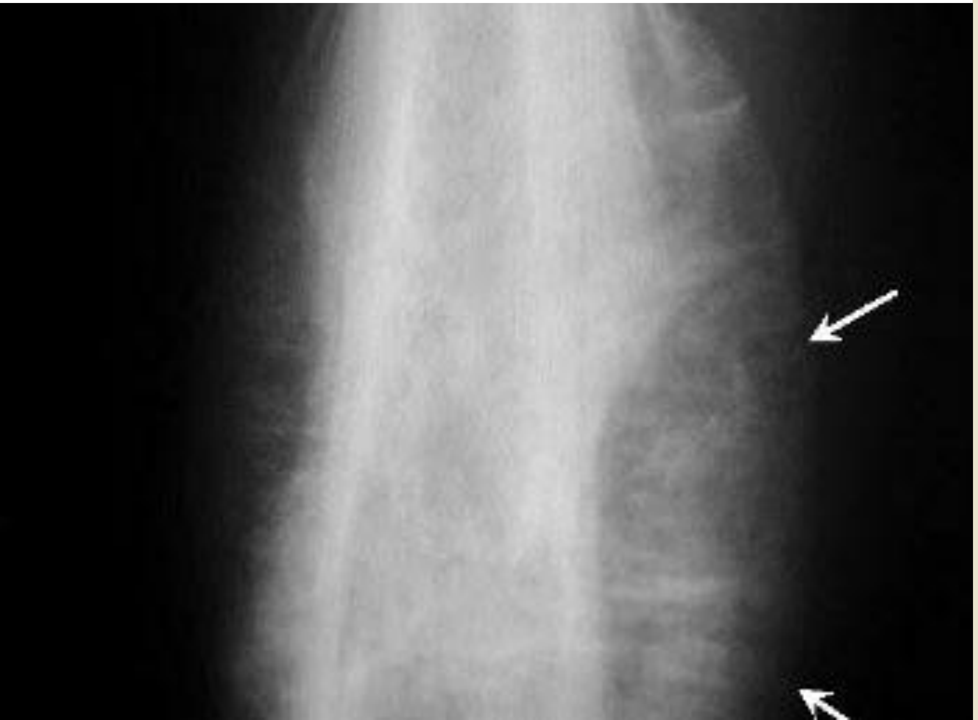
- a palpable mass,
- restricted joint motion,
- pain with weight bearing
- localized warmth/erythema.

Diagnosis and Staging

1. Radiology
2. Biopsy
3. Lab investigations

Radiology



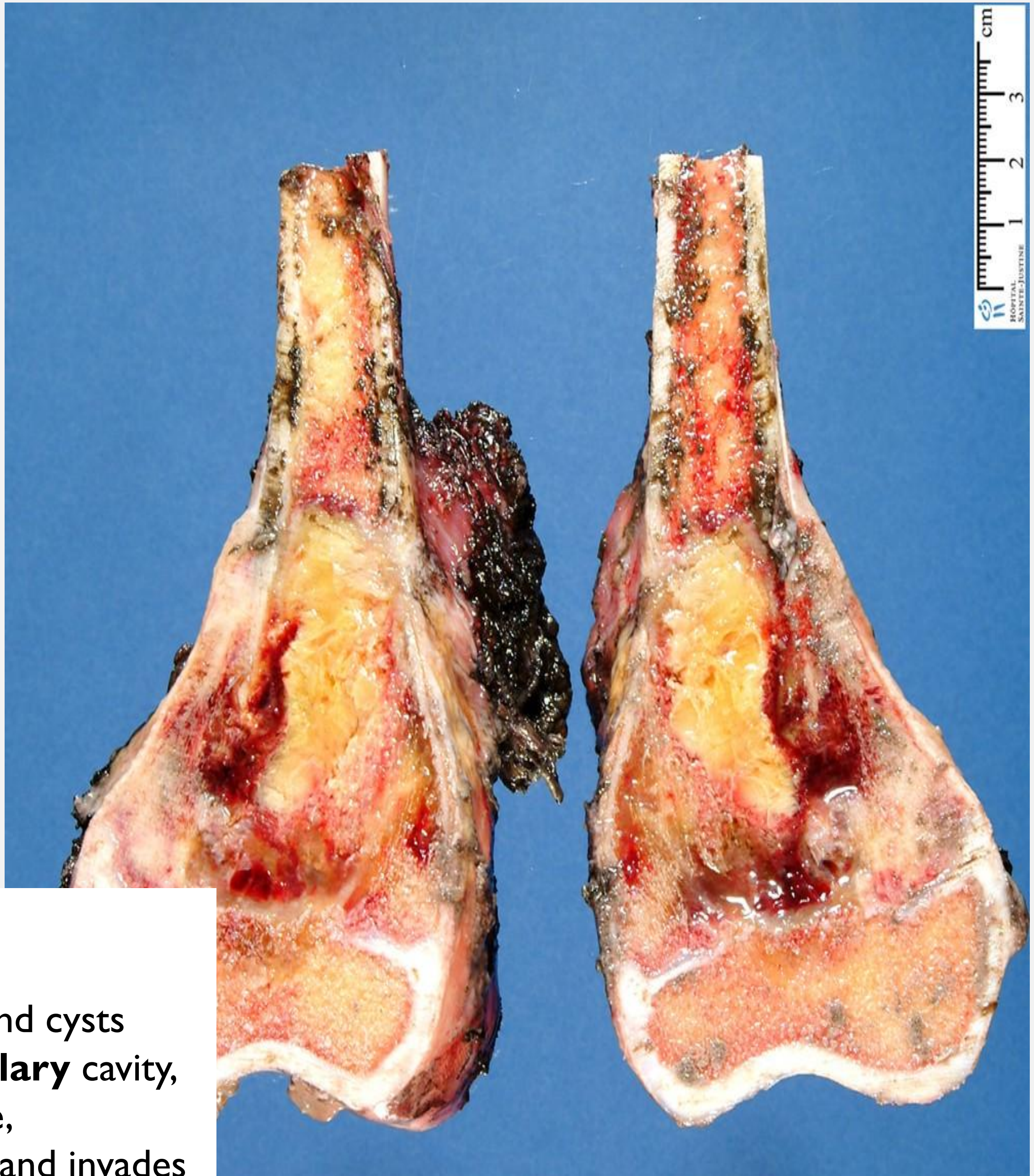


Large, ill defined, destructive, **lytic or blastic** mass

May break through **cortex** and elevate **periosteum** forming **Codman triangle (reactive bone)**

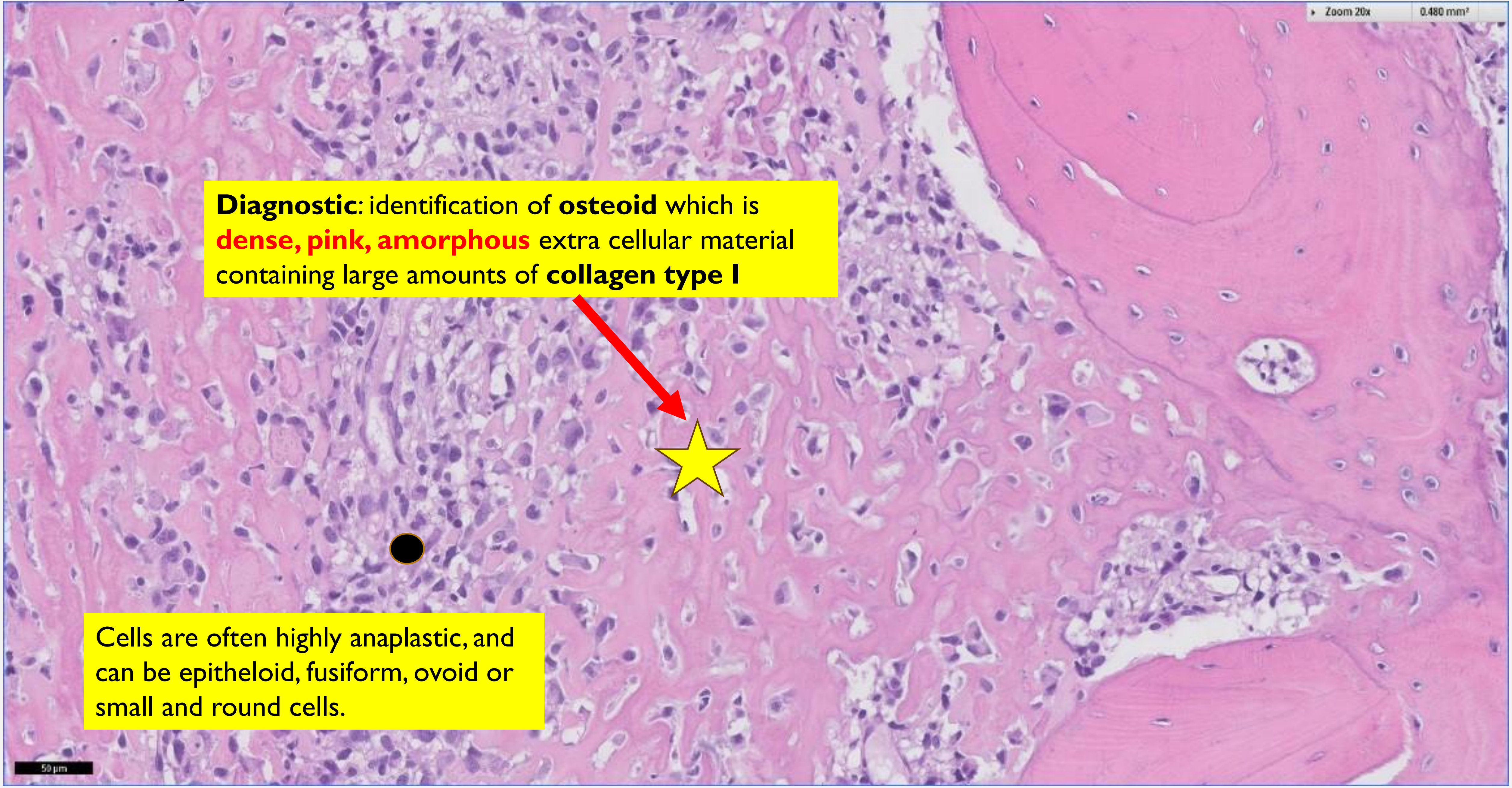
Sunburst pattern due to new bone formation in **soft tissue**

Gross

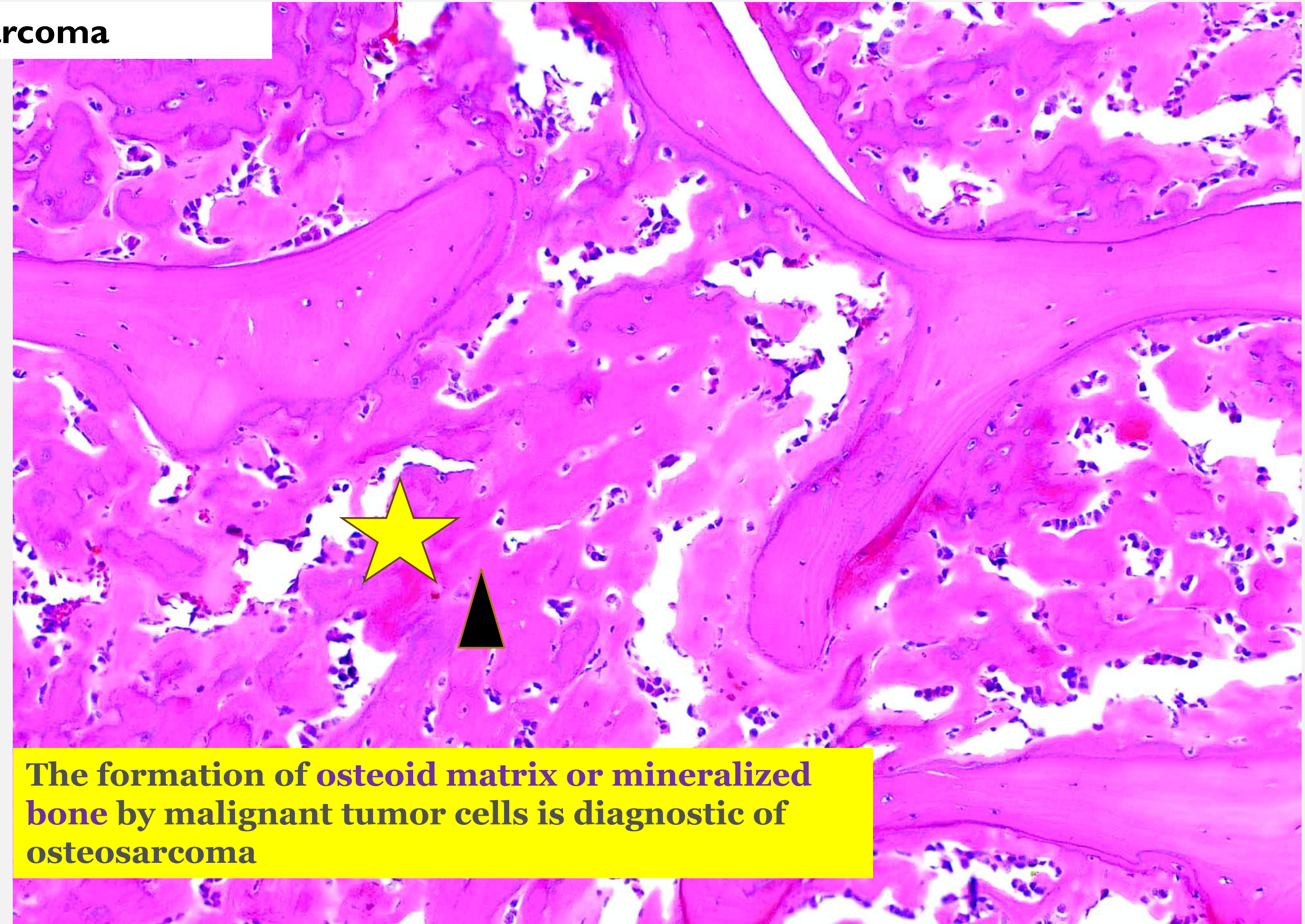


metaphysis
Big, bulky and gritty
Areas of hemorrhage and cysts
Spreads within **medullary** cavity,
destroys **cortical** bone,
elevates **periosteum** and invades

Microscopic



Osteosarcoma



The formation of osteoid matrix or mineralized bone by malignant tumor cells is diagnostic of osteosarcoma

Osteosarcoma